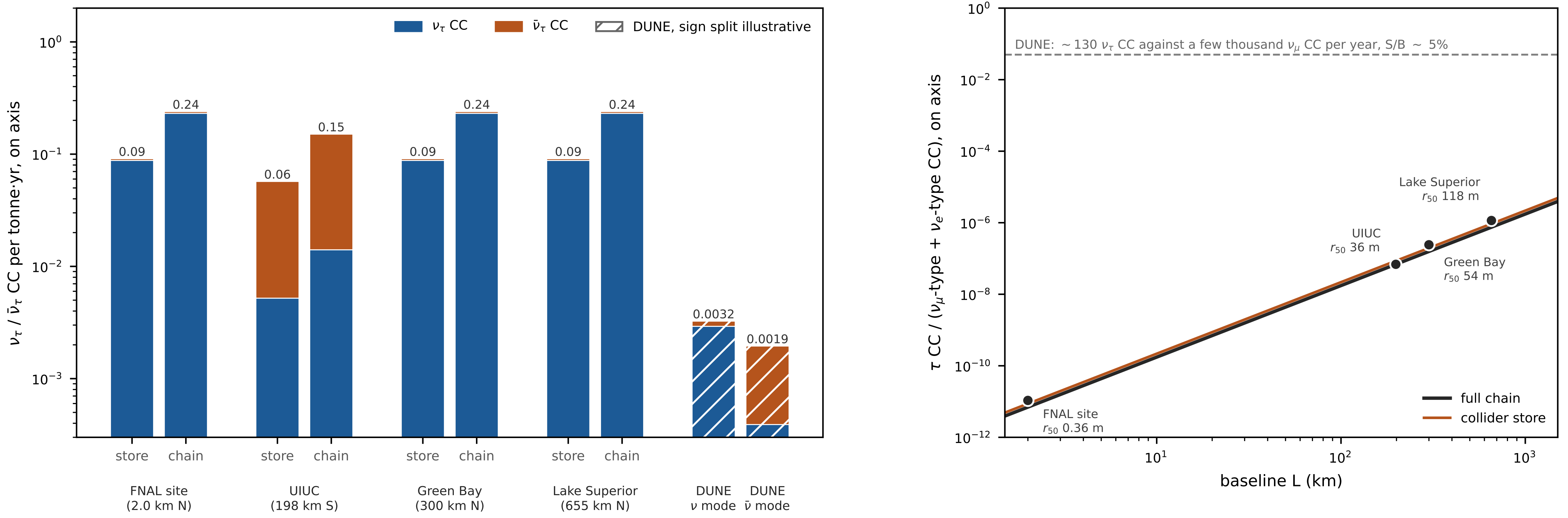
ν_τ versus $\bar{\nu}_\tau$ at four sites, per tonne-year, against DUNE

per tonne on the axis: the same at every far site; the SIGN is set by which way μ^+ circulates
(here μ^+ southbound on the east straights: UIUC gets $\bar{\nu}_\tau$, the northern sites ν_τ; the minority sign is $\nu_e \to \nu_\tau$)

what baseline buys: signal-to-background $\propto L^2$ (rate does not change); r_{50} = collider spot radius



Rates for a point detector on the beam axis (tau_compass.py machinery: north-aimed decays per cycle, 5 Hz, 1.2×10^7 s/yr, CSMS CC, τ threshold). $\nu_e \to \nu_\tau$ amplitude 0.050 vs 0.95 for $\nu_\mu \to \nu_\tau$.
At 2 km the collider spot is 0.36 m and the pencils' 4 cm to 3 m: a tonne cannot sit on the axis, and S/B is 10^{-11}. DUNE: TDR 130 ν_τ CC/yr in 40 kt (ν mode, before efficiency); $\bar{\nu}$ mode scaled by 0.6 from the TDR's background ratios.